

NELSON KANG

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Education

University of Waterloo

Bachelor's Degree, Computer Science and Finance (CFM)

Activities: Poker Club Executive, Competitive Intramural Volleyball Captain, Training for Ironman 70.3 Muskoka 2027

Experience

Backend & Data Engineering Co-op

Jun 2026 – Aug 2026

Aquanow

Toronto, ON

- Ported a legacy JavaScript monthly reconciliation job to a **Lakeflow** worker (**Kafka**-triggered, containerized) over **Apache Iceberg**, validating **3,800+** identities against legacy output. Root-caused **two long-existing production discrepancies** that had gone undetected.
- Led its expansion into an **AI-agentic reconciliation workflow** that classifies breaks by failure mode and fans out investigation on residuals. Directing a second co-op engineer, extracted a shared **recon-kernel**, proved it on a **second reporting domain**, and authored the org-level **RFC and ADR** for scheduled alerting.
- Diagnosed and fixed **5 critical production bugs** in a live regulatory reporting system (**Python, AWS S3, Athena, PyIceberg**) submitting daily trading activity to **Turkey's national regulator (MKK)**, including a full-day snapshot-boundary offset and in-asset fee misattribution. Restored reconciliation from **0% to 107/107 currencies**.

Software Engineer Intern

Jun 2024 – Aug 2024

Ericsson Canada Inc.

Ottawa, ON

- Built **4 ETL pipelines** using **C# and SQL Server** to extract Jira data from JSON payloads, transform custom field mappings, and **join across multiple Jira endpoints and internal tables** to load into reporting tables; used **temporal tables** for full change-history tracking.
- Audited and catalogued all existing team pipelines, identifying and documenting faulty custom-field mappings. Created a reference spreadsheet that became the team's **pipeline debugging resource**.
- Developed and maintained **Power BI dashboards** for trouble report visibility, supporting stakeholder reporting and data-driven decision-making across the BI team.

Projects

ClipFarm — Automated Volleyball Highlight Generator

Mar 2026 – Present

Python, FastAPI, Next.js, TypeScript, YOLOv8-pose, Roboflow, PostgreSQL, Docker, Cloudflare R2

- Initiated and lead a team of 4 engineers** building a **multi-signal highlight pipeline**: a custom-trained **Roboflow** ball detector tracks trajectory to bound rallies, **audio cheer detection** and rally-shape scoring gate highlights, and **YOLOv8-pose** classifies player actions.
- Containerized with **Docker Compose**: **FastAPI** serves auth-gated endpoints, **Celery + Redis** distributes jobs, and **Modal** offloads GPU inference. Auth via **Supabase JWKS** with **Row-Level Security**.
- Implemented the frontend in **Next.js + TypeScript** with a filterable clip grid, inline multi-label correction UI for future model training, clip trimming with background re-cut via Celery, and lazy-loaded thumbnails.
- Established the team's engineering process: **Agile** sprints over a scoped issue board, **protected main**, and **CI gates** (ruff, mypy, pytest, ESLint) on every merge. Built an **autonomous overnight AI agentic loop** that clears non-blocking tickets, opens PRs with multi-pass automated review, and reports each morning for human sign-off.

HUME — 1st Place, CORTEX BioSciences Track @ Hack the 6ix (\$1,500 + work trial)

Jul 2026

Python (stdlib only), Knowledge Graphs, Adversarial Robustness, Metamorphic Testing

- Built a **deterministic belief-revision engine (no LLM)** that ingests streaming evidence into a **knowledge graph**, sizing each update in **log-odds** by evidence strength and scoping refuted claims to the condition that breaks them.
- Blocked **prompt-injection** structurally. Untrusted text cannot mutate state; writes pass a closed **delta vocabulary** with magnitude from **structured provenance** only. Independent **~90-attack red-team: zero** text-driven mutations.
- Locked extraction behind one audited seam with **19 probe suites: metamorphic tests** (paraphrase, entity renaming, negation) plus an **AST build guard**, catching two silent false-negative regressions.

Multi-Factor Portfolio-Advisor — 1st Place, Market Beat Competition

Nov 2025

Python, Pandas, NumPy, yfinance

- Optimized a **\$1M portfolio** achieved **5.7% return** vs. 3.7% market average benchmark, placing **1st out of 18 teams**.
- Implemented a **multi-factor portfolio construction pipeline** applying momentum, volatility, correlation, and trend signals inspired by the **Ranked Asset Allocation Model (RAAM)**.
- Automated end-to-end: data ingestion, liquidity screening, factor ranking, **sector-cap enforcement**, and **Sharpe-ratio optimization** under real-world constraints (transaction fees, FX conversion, position limits).